

Part 1 — Go-Back-N

Open the Go-Back-N animation:

https://media.pearsoncmg.com/ph/esm/ecs_kurose_compnetwork_8/cw/content/interactiveanimations/go-back-n-protocol/index.html Links to an external site.

Run the animation multiple times. Pause it. Try introducing packet loss.

Answer the following:

1. When a packet is lost, what does the receiver send back?
The receiver sends nothing back because they never got the packet to respond to.
2. What event causes the sender to retransmit packets?
If the receiver gets a packet but not the one before it, it will retransmit the packets from the lost packets to the current packet. So for example, if I send packets 2 through 5 but packet 3 is lost, packets 3,4,5 will be retransmitted in that order to the receiver.
3. If you send packets 0-4 out, and packet 2 is lost, which packets end up being retransmitted?
Packets 2,3,4 will be retransmitted.
4. Does the receiver buffer out-of-order packets? What behavior do you observe?
No. If it gets packet 4 but wants 3, it drops 4 and sends an ACK for 2.
5. Why might this protocol retransmit packets that were actually delivered correctly?
Because the receiver refuses to hold them. To the sender, if 3 timed out, everything after it is "lost" as far as the receiver's window is concerned.
6. In what kind of network conditions (low loss vs high loss) would Go-Back-N perform reasonably well? Explain.
Low loss. It works fine when errors are rare, but in high loss, it wastes too much time re-sending things the receiver already saw but threw away.
7. What makes Go-Back-N relatively simple to implement?
The receiver only has to track one number for the next packet it wants. It doesn't need to manage a big memory buffer or sort anything.

Part 2 — Selective Repeat

Open the Selective Repeat animation:

https://media.pearsoncmg.com/ph/esm/ecs_kurose_compnetwork_8/cw/content/interactiveanimations/selective-repeat-protocol/index.html Links to an external site.

Run it multiple times. Introduce packet loss. Observe carefully.

Answer the following:

8. If you send packets 0-4 out, and packet 2 is lost, which packets end up being retransmitted?
Only packet 2.
9. Does the receiver accept and store out-of-order packets? What do you observe?
Yes. It saves packets like 3 and 4 in a buffer and waits for 2 to show up to fill the gap.
10. What additional responsibility does the receiver now have compared to Go-Back-N?
It has to buffer and reorder the packets so they are in the right sequence before giving them to the app.
11. What kind of state must the receiver keep track of?
It has to track the status of every single packet in its window to know which ones are missing and which ones are already stored.
12. Why does Selective Repeat use bandwidth more efficiently under packet loss?
It only sends the specific packet that actually got lost instead of re-sending a whole bunch of successful ones.
13. What makes Selective Repeat more complex than Go-Back-N?
The sender needs a timer for every individual packet, and the receiver needs extra logic and memory to handle the buffer.

Part 3 — Direct Comparison

Properties:

Sender complexity: Go-Back-N because only it needs one timer for everything.

Receiver complexity: Go-Back-N because no sorting or buffering is needed.

Memory requirements: Go-Back-N because it doesn't need to store packets out of order.

Bandwidth efficiency under loss: Selective Repeat because it doesn't re-send data that already arrived.

For each property, indicate whether Go-Back-N or Selective Repeat has the advantage, and briefly explain why.

Final Reflection

14. If you were designing a protocol for:

- A high-latency satellite link with occasional packet loss
- A small embedded device with very limited memory

For a high-latency satellite link, I would pick Selective Repeat because you don't want to waste time re-sending an entire window of data just because one packet dropped. For a small embedded device with limited memory, I would choose Go-Back-N because it is much simpler and doesn't need extra RAM to buffer out-of-order data.

Which protocol would you choose in each case and why?

15. Based on what you observed, which protocol would scale better in a real-world Internet environment? Explain your reasoning.

Selective Repeat. Since the internet is fast and has lots of data moving at once, GBN would cause too much extra traffic by re-sending huge chunks of data every time one little packet drops.